

Experiment 5: Count Number of Occurrences in a Data Stream

Aim

To count the number of occurrences of each element in a data stream using single pass processing, maintaining a running count with a dictionary and displaying the final frequency of every element.

Theory

In stream processing, data arrives continuously and each element can normally be read only once. Because of this, algorithms must complete their work in a single pass over the data. Counting the number of occurrences is one of the most basic stream operations. A dictionary is used to store the count, where the key is the element and the value is its frequency. When an element arrives, if it is already present in the dictionary its count is increased by one, otherwise it is inserted with a count of one. The count is therefore always up to date and no second scan of the stream is required.

Requirements

- Python 3.x
- Jupyter Notebook

Step 1: Define Streaming Data

```
stream = ['A', 'B', 'C', 'D', 'A', 'B', 'C', 'D', 'E', 'A']
```

Interpretation: Defines the incoming data stream. Each element represents one arriving record.

Step 2: Initialize Counter

```
count = {}
```

Interpretation: Creates an empty dictionary that will store each element as a key and its frequency as the value.

Step 3: One Pass Processing

```
for item in stream:
    if item in count:
        count[item] += 1
    else:
        count[item] = 1
    print("Current counts:", count)
```

Interpretation: Reads every element once. If the element already exists its count is increased, otherwise it is added with a count of one. The running count is printed after every arrival.

Step 4: Display Final Counts

```
print("\nFinal Counts")
for key, value in count.items():
    print(key, ":", value)
```

Interpretation: Displays the final frequency of each element in the stream.

Complete Program

```

stream = ['A', 'B', 'C', 'D', 'A', 'B', 'C', 'D', 'E', 'A']

count = {}

for item in stream:
    if item in count:
        count[item] += 1
    else:
        count[item] = 1
    print("Current counts:", count)

print("\nFinal Counts")
for key, value in count.items():
    print(key, ":", value)

```

Sample Input

```
stream = ['A', 'B', 'C', 'D', 'A', 'B', 'C', 'D', 'E', 'A']
```

Expected Output

```

Current counts: {'A': 1}
Current counts: {'A': 1, 'B': 1}
Current counts: {'A': 1, 'B': 1, 'C': 1}
Current counts: {'A': 1, 'B': 1, 'C': 1, 'D': 1}
Current counts: {'A': 2, 'B': 1, 'C': 1, 'D': 1}
Current counts: {'A': 2, 'B': 2, 'C': 1, 'D': 1}
Current counts: {'A': 2, 'B': 2, 'C': 2, 'D': 1}
Current counts: {'A': 2, 'B': 2, 'C': 2, 'D': 2}
Current counts: {'A': 2, 'B': 2, 'C': 2, 'D': 2, 'E': 1}
Current counts: {'A': 3, 'B': 2, 'C': 2, 'D': 2, 'E': 1}

Final Counts
A : 3
B : 2
C : 2
D : 2
E : 1

```

Interpretation

- Stream elements arrive sequentially.
- Each element represents incoming data.
- A dictionary stores the occurrence count.
- Key = element.
- Value = frequency.
- Each item is processed once.
- The count updates immediately.
- No second scan is required.
- It shows the total occurrences of each element.
- A final frequency summary is generated.

Result

The number of occurrences of each element in the data stream was successfully counted using single pass processing. The final frequency summary was generated as A = 3, B = 2, C = 2, D = 2

and $E = 1$.